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White Paper

A Randomized, Double-Blind, Placebo-Controlled, Multi-Center Study to Evaluate the Safety and Efficacy of Eculizumab in Subjects With Refractory Generalized Myasthenia Gravis (gMG)

Introduction

Myasthenia gravis (MG) is a chronic autoimmune antibody-mediated disorder, characterized by defective neuromuscular synaptic transmission and the loss of acetylcholine receptors [1, 2]. The clinical hallmark of MG consists of fluctuating fatigability and weakness affecting muscle groups [3]. Conventional treatment strategies for generalized MG (gMG) include symptomatic therapy with acetylcholinesterase inhibitors, long-term immunosuppressive therapy with corticosteroids or steroid-sparing agents, and short-term interventions such as intravenous immunoglobulin (IVIG) or plasma exchange (PLEX) for acute exacerbations [4]. Refractory myasthenia gravis is generally defined as when a patient with MG fails to respond to conventional treatment adequately or frequently has myasthenic crises while undergoing therapy. Immunosuppressive drugs and steroids prove to be ineffective in treating this disease [5, 6]. Also patients with refractory MG are more likely to face life-threatening events and hospitalisations. Therefore, it is imperative to find an alternate treatment for patients suffering from this condition [7].

Mechanisms of action

Usage of eculizumab to treat symptoms of gMG is based on previous findings of membrane attack complexes (MACs) present in damaged tissue of MG patients [8]. The creation of MACs are reasoned to be due to complement activation by MG autoantibodies. This is further supported by the presence of IgG and C3 proteins at the same sites [8]. As the IgG autoantibodies bind the cell membranes of the neuromuscular junctions, C1 is recruited which in turn starts the classical pathway of complement activation, ending with C3b bound to the cell surface [9]. C3b will then facilitate the cleavage of C5 which in turn will lead to the creation of MACs. MACs create pores in the cell membrane that disrupt the equilibria present within the cell leading to cell death [10].

Eculizumab has a high affinity to uncleaved C5. When bound, eculizumab prevents the cleavage of C5 into C5a and C5b, which in turn stops the creation of C5b induced MACs [8]. Therefore, the usage of eculizumab is theorized to reduce tissue damage in gMG patients. Additionally it is suggested that eculizumab might be effective in treating patients with anti-acetylcholine receptor antibody-positive rMG [11].

The REGAIN study

To assess the safety and efficacy of eculizumab for refractory gMG, the randomized, double-blinded, placebo-controlled phase 3 REGAIN trial and its open-label extension were conducted [11].

Between April 2014 and February 2016, 170 patients were screened, 126 were randomly assigned to treatment and 125 received the study drug (62 eculizumab, 63 received placebo). Patients (18 years or older) with anti-acetylcholine receptor antibody-positive refractory gMG confirming gMG, MG-ADL $\geq$ 6, and MGFA class II–IV disease were eligible [11]. Of the 125 treated, 118 completed the study and seven discontinued, mostly for adverse events [12].

The primary objective of the study was to observe a statistically significant difference of the MG-ADL score between treatment and placebo groups from baseline to week 26. The statistical significance was determined with a worst-rank ANCOVA and it did not achieve significance between the eculizumab and placebo groups [11]. To provide a comprehensive clinical picture reflecting both the physicians' and patients' perspectives, prespecified sensitivity and secondary analytical approaches were included. The secondary objectives were evidence of overall safety and tolerability of eculizumab [12], as well as baseline difference in the QMG score, a responder analysis of the MG-ADL and QMG scores, a baseline difference in the MGC scale, and a baseline difference in the MG-QOL15 questionnaire. MG-ADL, MGC, and QMG reflect functional abilities and disease burden, while MG-QOL15 focuses on patient experience [11].

Eculizumab showed a benefit over placebo for QMG change, and also for the MG-ADL and QMG responder analysis. MG-QOL15 also favoured eculizumab, whereas worst-rank ANCOVA for MGC did not differ between groups. [11]. Sensitivity analyses for MG-ADL, QMG and MG-QOL15 showed improvements with eculizumab, and MGC results were consistent with MG-ADL [11].

In sum, even though the primary endpoint did not achieve a statistical significance, secondary and sensitivity analyses indicate eculizumab was well tolerated, with a safety profile consistent with prior diseases (PHN, aHUS).

Limitations of the study

The main limitation of the REGAIN study is the worst-rank analysis used as a statistical approach by the researchers. It enabled patients who needed temporary rescue medication to be included in the efficacy analysis by treating rescue medication use or discontinuation for any reason as a negative outcome. For this reason, the derived use of ranks rather than actual change scores in the

worst-rank analysis makes it difficult to translate the implications of the study results into clinical practice [11].

Despite the efficacy of eculizumab in treating AchR antibody-positive refractory gMG was demonstrated in the REGAIN study, there is little information on its efficacy in other subgroups of MG patients including seronegative MG, thymoma-associated MG and MG crisis [13]. More research into these groups will be really interesting to develop effective treatments against the disease.

Stakeholders Perspective

Patients' Perspective

REGAIN has shown eculizumab does not prove a significant risk in comparison to other immunosuppressant medications and that there's an improvement of perceived fatigue and an alleviation of symptoms. This means that the patient does not considerably increase the already present risk posed by the standard immunosuppressants and they have an overall better well being. Therefore usage of eculizumab is for the patients' benefit. The scoring of MG-QOL15 agrees with patients feeling better when on eculizumab [13, 14].

Myasthenia gravis incidence increases with age, peaking at 30 and 50 years of age in women and 60-89 years of age in men. Women are three times more likely to have early onset MG; defined as being affected before 40 years of age [1]. Succumbing to the disease before 40 means that there are at least 20 predicted years of work being lost to some degree. This will affect the patients' economic situation and there is also a societal impact from losing many working years from the early onset MG patients. The more female dominated occupations will be more affected by this as the ratio is higher for women succumbing to the disease earlier in life. The peak in male incidence is around retirement age, which is not as big of a loss for society from the labour perspective.

There have been studies indicating a phenotype difference of MG dependent on ethnic background [1]. However treatment efficiency of eculizumab has been shown to be similar between japanese and caucasian patients, which were compared due to their differences. Therefore the treatment of the MG pathology that requires the use of eculizumab should not be affected by ethnic background [15].

Pharmaceutical Companies' Perspectives

From the perspective of the pharmaceutical company, the pricing of Eculizumab is mainly considered by the clinical value, the R&D risk, and the rarity of the market. Due to the limited number of patients, the company needs to ensure the R&D will get reasonable payback, in the

meanwhile, the price also reflects the compensation of innovation for the high risks of development [16].

Despite the opportunities, the eculizumab market faces significant challenges. Regulatory hurdles remain a concern, as the approval processes for new indications and formulations can be lengthy and complex [17]. The Food and Drug Administration (FDA) and other regulatory agencies enforce strict compliance measures that may delay market entry, impacting the overall growth potential.

Apart from this, the appearance of a biosimilar threatens eculizumab. Even with a patent, a biosimilar drug can enter the market, which will aggravate pricing. Influencing the revenues for the originator product. A report by the FDA highlighted that biosimilars could lead to cost savings of up to 30%, which could make eculizumab less accessible in the long term [18].

Caregivers' and Patient Relatives' Perspective

The burden of gMG is not only limited to the patient, but also has a profound influence on the family. The expense of daily life, decreased income, and the need to balance economic needs must also be considered [19].

The caregiver has to take responsibility for caring, giving up the opportunity of a job chance. Additionally, the family needs to have long-term planning, to ensure the uncertainty of the chronic disease treatment (e.g. the coverage of insurance). The therapy may be interrupted, so the family needs to have a contingency plan, such as complement inhibitors, which has the potential to not only improve patient outcomes but also relieve caregiver burden [20].

The efficiency of eculizumab improves the quality of family life by decreasing attacks and the number of hospitalizations. However, the continuous high treatment cost means there should be a balance between the hope of improvement and the sustainability and accessibility of long-term treatment.

Hospitals' Perspective

Eculizumab is administered through regular intravenous infusions. This requires dedicated infusion facilities, nursing staff, and monitoring protocols. For hospitals, this translates into increased demand on infusion suites and staff time compared to oral therapies [21].

On the other hand, the severe myasthenic crises are reduced with the treatment, translating this into less admissions in Intensive Care Units, a lower need of mechanical ventilation and less rescue

treatments. Moreover, a reduction in the use of corticosteroids and immunosuppressants imply less derived complications (osteoporosis, infections), which reduces as well the number of hospitalizations [22].

Organisational benefits include the optimization of hospital resources, scheduled outpatient treatment and a better adherence and follow-up. Strategic benefits can even be reported as the hospital can be positioned as a referent in rare neuromuscular diseases due to its access to innovative therapies, which can provide scientific visibility and the participation in research projects [22].

Ethical Considerations

Costs

The cost of eculizumab is among the highest of any biologic therapy, often exceeding USD 400,000–500,000 per patient per year. For hospitals operating under budgetary constraints, this creates tension between providing cutting-edge therapy and ensuring equitable access to care for the broader patient population. In systems where hospitals share financial responsibility, high-cost treatments can strain institutional budgets, potentially limiting investment in other essential services [23].

Cost-effectiveness should be considered besides the efficacy and tolerability when choosing a drug. There are no pharmacoeconomic analyses of eculizumab in patients with gMG currently. Well-designed studies assessing the cost-effectiveness of eculizumab are needed so as to reduce the expensive financial burden of eculizumab on MG patients [13].

In addition, defining optimal treatment duration and maintenance dose, seeking biomarkers of activity and response to treatment, combining with other immunosuppressant therapies are needed for the better control of medication cost in MG patients [13].

As there is no in depth study done, we can only make an estimate based on the information currently available. Eculizumab has been designated as an orphan drug [24], therefore governments believe that access to treatment using eculizumab for MG, and for other rare diseases, is of enough importance to receive monetary support [11]. This could be due to the major financial loss to society should 20+ years of active work from several individuals be lost. As the treatment also reduces the need for rescue care there is also a financial gain in saving hospital resources [22]. Therefore the cost of the treatment seems to be a necessity to reduce a larger cost in loss of labour.

Legislation

Treatment of gMG using eculizumab was approved for use in the EU by the European Medicines Agency (EMA) in 2014 [24]. A second treatment option using a peptide called zilucoplan working in similar ways to eculizumab was accepted by Läkemedelsverket in 2023 [25]. To be eligible for treatment with eculizumab, patients need to be vaccinated against *Neisseria meningitidis*, this is due to C5 inhibitors having a known infection risk associated with *N. meningitidis* [24, 25].

Risks

The most common adverse events are headache, nasopharyngitis, arthralgia, diarrhea and infections, most of them being mild to moderate symptoms. Serious infections such as aspergillus and pseudomonas were also reported, but direct relationship with eculizumab cannot be definitely established considering infections may occur in other immunosuppressive therapies [13, 26, 27].

In REGAIN, the incidence of treatment-emergent adverse events were generally similar comparing with placebo, and most adverse events were mild to moderate in severity, just an extreme case reporting one death of a patient due to crisis-related complications [13].

In the open-label extension study, there were no occurrences of meningococcal infection by the end of the trial, only one non-fatal case was reported afterwards. The drug showed a good tolerability and no specific safety concerns were found [12, 13]. Therefore, eculizumab appears to be substantially safe in the long-period treatment of MG.

Summary and Recommendations

The REGAIN study provides strong evidence supporting the use of eculizumab in patients with gMG unresponsive to conventional therapies. It has demonstrated to improve muscle strength, functional ability and quality of life, while maintaining a favorable safety profile. Compared with existing treatments, eculizumab minimizes disease exacerbations, hospitalizations, and long-term reliance on immunosuppressants, lowering the burden for patients, relatives, caregivers and healthcare systems.

While the clinical benefits of eculizumab are well established, it remains essential to outline how these findings can be translated into practice. While it requires comprehensive cost-effectiveness studies against current treatments, they should reflect why the overall cost to the healthcare system may be lower in the long run. It should also be ensured that the drug gets official authorization for use and it is financially accessible through healthcare systems, guaranteeing a fair access to everyone. Putting into place all these measures would help transform evidence into actual clinical use and ensure the therapy reaches patients in a sustainable and equitable way.

Based on all the evidence, we strongly recommend the integration of eculizumab into the treatment of gMG, with further evaluation of its economic impact and equitable implementation strategies.

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Justification for the use of AI

The use of the AI ChatGPT in this White Paper is focused on grammar corrections, rewriting sentences for higher clarity, and obtaining relevant scientific references. The students' criteria is the first source for this work and AI serves as a tool to optimize scientific communication. Thus, AI is a complement to improve the formal quality of this paper and facilitates access to concrete literature, the scientific work presented is therefore made by the authors and not through AI.